

Pulseseq-CEST: Towards multi-site multi-vendor compatibility and reproducibility of CEST experiments using an open-source sequence standard

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Purpose: As the field of CEST grows, various novel preparation periods using different parameters are being introduced. At the same time, large, multisite clinical studies require clearly defined protocols, especially across different vendors. Here, we propose a CEST definition standard using the open *Pulseseq* format for a shareable, simple, and exact definition of CEST protocols.

Methods: We present the benefits of such a standard in three ways: (1) an open database on GitHub, where fully defined, human-readable CEST protocols can be shared; (2) an open-source Bloch-McConnell simulation to test and optimize CEST preparation periods in silico; and (3) a hybrid MR sequence that plays out the CEST preparation period and can be combined with any existing readout module.

Results: The exact definition of the CEST preparation period, in combination with the flexible simulation, leads to a good match between simulations and measurements. The standard allowed finding consensus on three amide proton transfer-weighted protocols that could be compared in healthy subjects and a tumor patient. In

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addition, we could show coherent multisite results for a sophisticated CEST method, highlighting the benefits regarding protocol sharing and reproducibility.

Conclusion: With *Pulseq*-CEST, we provide a straightforward approach to standardize, share, simulate, and measure different CEST preparation schemes, which are inherently completely defined.

KEYWORDS

CEST, open-source, *Pulseq*, standardization

1 | INTRODUCTION

Chemical exchange saturation transfer MRI uses the exchange transfer of magnetization from solutes to water to increase the sensitivity of detection through a saturation effect on the water signal.¹⁻³ The CEST method uses a molecular amplification mechanism that accumulates its effect on the spin system during a saturation period (T_{sat}), consisting of one or more RF pulses with or without interpulse delays, followed by an imaging sequence. Detectable CEST effects in vivo have been reported for instance for proteins,^{4,5} glutamate,⁶ and different sugars.⁷⁻⁹ The choice of a specific saturation period is crucial for an optimal CEST experiment, as the maximum effect depends not only on the tissue and solute pool of interest, but also on the efficiency of the saturation imposed by the RF pulse scheme¹⁰ and of its transfer during T_{sat} ^{11,12} as well as on concomitant saturation effects, such as direct saturation and magnetization-transfer contrast associated with the semisolid pool.^{13,14} Moreover, differences in approaches for data analysis in terms of normalization or spectral regions considered can further affect the final image contrast calculated from Z-spectra.¹⁵

Thus, the saturation period has to be precisely defined by such parameters as RF pulse shape (both magnitude and phase), pulse duration (t_p), saturation duty cycle (DC_{sat}), total saturation time (T_{sat}), saturation field strength (B_1), and offset from the water resonance frequency ($\Delta\omega$). However, these parameters vary significantly in the current literature,¹⁶ and are not always provided in sufficient detail. In addition, the literature uses different definitions or terminology to describe saturation “power,” such as the flip angle of the pulse,¹⁷ pulse peak B_1 amplitude,¹⁸ B_1 root mean square ($B_{1,\text{rms}}$) amplitude,¹⁹ or continuous wave power equivalent ($B_{1\text{cwp}}$),²⁰ potentially leading to confusion when implementing a comparable CEST MRI experiment. Therefore, it is not always possible to faithfully reproduce a CEST experiment without corresponding with the authors, and even then, the method could still be prone to errors. Thus, a common, easy-to-use format for researchers to provide and share the precise saturation parameters is desirable, especially regarding the current focus on reproducibility in MR research.²¹ Moreover,

a growing number of deep learning-based evaluation approaches for large multi-site, multi-vendor data sets make a proper definition of input data even more important.²²

A vendor-independent, human-readable, and sharable file format for MR sequences has been introduced with the *Pulseq* framework.²³ In *Pulseq*, all sequence parameters are defined in a text file (hereafter called *pulseq-file*), which can be created with various popular programming applications such as *MATLAB* (The MathWorks, Natick, MA) or *Python*.²⁴ This *pulseq-file* is then read and played out on the scanner using a vendor-specific interpreter sequence. Although *Pulseq* is a great tool that enables a flexible implementation of complex sequence patterns,²⁵ it is complicated to incorporate vendor-provided image reconstruction functions, which are generally proprietary. However, having the source code of a full interpreter sequence at hand, a capsulated interpreter can be included into other existing sequences for imaging readout. For example, a 3D snapshot gradient echo^{26,27} can be equipped with an encapsulated *Pulseq* interpreter that solely plays out a CEST preparation block defined in a *pulseq-file*. This makes the *Pulseq* file format a perfect candidate for sharing CEST preparation periods. The established MRI readout following the saturation period can be used with the familiar user interface and all possibilities of adjustments and image reconstruction. This procedure enables four major advantages:

1. The CEST preparation period definition in *Pulseq* is complete. It is defined in a human-readable text file that is easy to interpret and allows direct comparison of different protocols. Thus, exchanging and comparing such files allows total reproducibility;
2. The definition of the RF pulses can be done in *MATLAB* or *Python* instead of implementing it in the sequence using a vendor-specific language (often C++), which, depending on the vendor, can be time-consuming and can require the compilation of a new sequence library;
3. The CEST preparation period can be used directly in simulations in the same framework (eg, *MATLAB* or *Python*), eliminating possible sources of error from transferring simulation results to the sequence source code and vice versa; and

4. The CEST preparation period can be used directly at the scanner with different state-of-the-art readouts, bridging the gap from first publication to reproducible multi-site application not only for research, but also for clinical applications. In addition, novel developments and work-in-progress approaches can be compared much faster and more reliably than with existing approaches.

Using *Pulseseq* for the CEST preparation part in the sequence theoretically enables a vendor-independent approach, provided that a *Pulseseq* interpreter sequence is available for each vendor. In this work, we implemented such a hybrid *Pulseseq*-CEST sequence for the Siemens IDEA (Integrated Development Environment for Applications; Siemens Healthineers, Erlangen, Germany) framework and tested it on three Siemens scanners at three sites, with two scanner models running very different software baselines. In addition, we present a fast and flexible open-source simulation for the same *pulseseq-files* that are played out on the MR scanner. Moreover, we provide a platform for researchers to share and test their saturation protocols in the *Pulseseq* format. As a first illustration of these steps, we show applications at clinically available MR scanners including adiabatic spin lock-prepared imaging, well-defined and original-author-approved amide proton transfer-weighted (APTw) imaging, and a CEST MR fingerprinting (MRF) protocol measured at three different MR sites in Europe and the United States.

2 | METHODS

2.1 | Pulseseq to standardize CEST preparation periods

Originally intended as a hardware-independent MRI sequence prototyping framework, *Pulseseq* allows for rapid and simple sequence definitions from within *MATLAB*, *Python*, and other software programming packages, which are usually open source.^{23,24,28} Within these programs, RF pulse, gradient, ADC, and trigger events can easily be defined and are written to a *pulseseq-file*, which is then read and played out by a native interpreter sequence on the scanner. Because *Pulseseq* includes built-in functions to generate block, Gaussian, apodized sinc, and arbitrary user-defined pulse shapes, theoretically, every excitation or saturation CEST preparation scheme can be defined with only a few lines of code. Thus, defining saturation periods in *Pulseseq* is a general and easy approach. Example codes to create different CEST preparation periods in *MATLAB* and *Python* are provided at the projects' website (<https://pulseseq-cest.github.io>).

The *pulseseq-file* contains the full definition of all sequence objects, which makes it a perfectly suitable candidate for

sharing protocol parameters of preparation periods for generating different types of contrast before readout, such as a CEST contrast here. Moreover, these *pulseseq-files* are human-readable, and the aforementioned *MATLAB* and *Python* packages include plot functions to compare preparation schemes directly. The benefit of such a direct comparison becomes obvious in Figure 1. Here, RF magnitude and phase of different CEST preparation periods used in this study are shown. For instance, Figure 1A,B shows the shape of an off-resonant adiabatic spin-lock pulse scheme previously used for in vivo DGEp studies²⁹ and phantom measurements herein. While such a pulse shape would be rather complex to describe in a publication, it is completely defined in the *pulseseq-file*. Figure 1C-H shows three different saturation preparation protocols (APTw_3T_001,³⁰ APTw_3T_002,³¹ and APTw_3T_003³²) that were recently recommended for APT-weighted tumor applications.³³ For comparison of the CEST preparation pulses, we use the following definitions for the B_1 pulse average ($B_{1,pa}$), B_1 average amplitude over pulse train ($B_{1,cwae}$), and B_1 average quadratic amplitude over pulse train ($B_{1,rms}$ or $B_{1,cwpe}$)^{34,35}:

$$B_{1,pa} = \frac{1}{t_p} \int_0^{t_p} B_1(t) dt \quad (1)$$

$$B_{1,cwae} = \frac{1}{t_p + t_d} \int_0^{t_p} B_1(t) dt \quad (2)$$

$$B_{1,rms} = \sqrt{\frac{1}{t_p + t_d} \int_0^{t_p} (B_1(t))^2 dt} \quad (3)$$

where t_p is the pulse duration, and t_d is the delay between preparation pulses. For the three different APTw protocols in Figure 1C-H, $B_{1,rms}$ was set to 2 μ T. The different peak amplitudes of the saturation pulses due to the different shapes and duty cycles are directly observable. For instance, in Figure 1C,E, sinc-Gauss pulses with the same shape are shown, but in Figure 1E, the peak amplitude of the pulse is higher, as this protocol uses a saturation duty cycle (DC_{sat}) of 0.5 compared with 0.9 for the protocol in C.

Figure 1I,J shows the RF magnitude and phase over multiple repetitions of the CEST-MRF schedule.³⁶⁻³⁸ Here, the amplitude of the spin-lock saturation pulses changes over the different repetitions.

Full details about the protocols can be found in the *pulseseq-files* in the Supporting Information. In addition, the RF phase evolution during saturation pulses is available, a parameter that is rarely provided in CEST literature, although it can have an influence on the experiment, as shown in more detail in Supporting Information Figures S1 and S2.

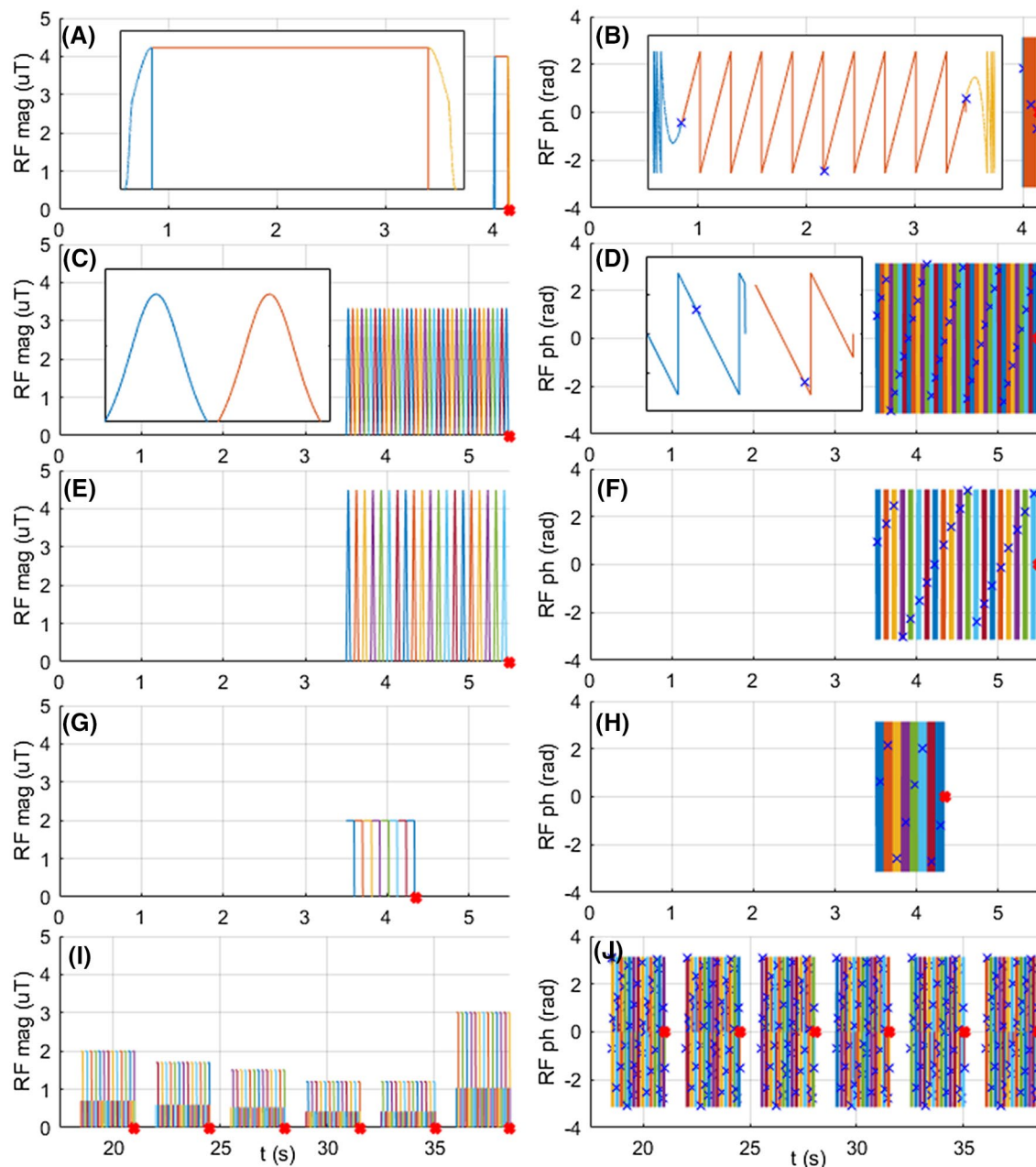


FIGURE 1 A,B, The RF magnitude and phase of the HSExp spin-lock saturation at 0.6 ppm. Adiabatic tip-down (blue) and tip-up (yellow) pulses surround the locking pulse (red). The frequency modulation can be seen in the zoomed plot (black box) from the changing phase. The red dot marks the beginning of the readout period, which can be an FID or a full 3D gradient echo for instance; spoiler gradients are not shown for simplicity. The RF magnitude (C,E,G) and phase (D,F,H) during three different amide proton transfer-weighted (APT_w) protocols: APT_w_3T_001 (C,D), APT_w_3T_002 (E,F), and APT_w_3T_003 (G,H)—all with $B_{1,\text{rms}} = 2 \mu\text{T}$ and recovery time ($T_{\text{rec}} = 3.5$ seconds). Due to the different pulse shapes and DC_{sat} during the preparation periods, the peak amplitudes of the pulses differ. In (C) and (D), a zoomed graph for two RF pulses is shown in the black rectangles. The phase accumulation due to the off-resonance character of the pulses is taken into account, as shown between the blue and red phase curve. The blue “x” marks the phase at the center of the RF pulse. I,J, The RF magnitude and phase of multiple repetitions during an amide-MR fingerprinting spin-lock saturation at 3.5 ppm

2.2 | Bloch-McConnell simulations

To be able to not only compare protocol parameters, but also simulate them, an application was written in C++ that loops through the *pulseseq-files*, performs Bloch-McConnell

simulations for the respective sequence events,³⁷ and returns the current magnetization vector after preparation. The compiled code is callable as a *mex* function for an easy integration into a *MATLAB*-based pipeline. The *Python* implementation wraps the C++ code with SWIG (Simplified Wrapper and

Interface Generator) and is simplified by a *Python* parser. This setup ensures input and output compatibility between the *MATLAB* and *Python* implementation. Where not specified explicitly, the *mex* function was used for the simulations in this study. Due to the flexible design, it is possible to simulate an arbitrary number of CEST pools and an additional semisolid magnetization-transfer contrast pool with either a Lorentzian or super-Lorentzian line shape. In combination with the flexible *Pulseseq* saturation period definition, any number of CEST pools can be simulated for any kind of preparation period in a relatively short amount of time, thanks to the native C++ implementation. For this study, the simulation program was compiled for a 64-bit *Microsoft Windows* 10 OS, using the *Microsoft Visual C++* 2017 compiler. Simulations were performed on a PC with an Intel i7-7700K Kaby Lake CPU. The source code is available on the project's website (<https://pulseseq-cest.github.io/>).

2.3 | Pulseseq sequence building block

By adapting the source code of the original *Pulseseq* sequence for the Siemens IDEA framework, we were able to play out the *pulseseq-files*, containing the definition of the CEST preparation period, directly on the scanner, followed by different readout sequences. Adaptions were implemented to use the code as a so-called sequence-building block (SBB). For example, (1) the FOV positioning options are removed, as this information is defined in the readout sequence; and (2) during the *Pulseseq* building block, no data are acquired, as indicated by the ADC event; instead the ADC event is used internally as marker to interrupt the *Pulseseq* sequence and switch to the readout sequence, which is then played out (Figure 2). Note that the *Pulseseq* block containing an ADC event is skipped entirely, and therefore is not allowed to contain any other events. With this design of the *Pulseseq* SBB, it can be implemented in every MR sequence where the source code is available. Because timing and specific absorption rate calculations are handled by the interpreter sequence, the workload for such implementation is small. Only a few lines of code need to be implemented in the main sequence for initialization, preparation, and running of the *Pulseseq* SBB. For instance, to implement the SBB to an established, native EPI sequence,^{39,40} only 30 lines of C++ code were necessary, including code regarding communication with the user interface.

The *Pulseseq* interpreter sequence for Siemens IDEA contains vendor-specific code, and can therefore only be obtained through the customer-to-customer partnership program (so-called C2P procedure). The interpreter sequence can be provided upon request to interested researchers. The current *Pulseseq* C2P package supports multiple software and hardware platforms of Siemens, including Numaris4 vb15, vb17,

vb19(a,b), vd11d, ve11(a,b,c,e,k,u), ve12u, and NumarisX va11 and va20. For other vendors, different “interpreter” approaches need to be developed or already exist, such as the TOPPE interpreter for GE (GE Healthcare, Chicago, IL) systems.⁴¹

2.4 | Phantom preparation

A phantom was prepared using multiple 6-mL tubes with either L-arginine or D-glucose. Five tubes were filled with 50 mmol/L L-arginine (Fluka Chemie, Buchs, Switzerland) dissolved in phosphate-buffered saline (according to Cold Spring Harbor Protocols,⁴² but containing 2.7 mM KCl, 10 mM Na₂HPO₄, 1.8 mM KH₂PO₄, and 140 mM NaCl). The pH of the L-arginine tubes was adjusted between 4 and 6 using HCl (Sigma-Aldrich Laborchemikalien, Germany) and NaOH (Riedel-de Haën, Seelze, Germany). Two additional tubes were filled with a D-glucose solution (Glucosteril 20%; Fresenius Kabi Deutschland, Germany) and diluted to 33 mmol/L or 66 mmol/L using phosphate-buffered saline. To shorten T₁, gadoterate meglumine (Dotarem 500 mmol/L; Guerbet, Villepinte, France) was added to each tube in this phantom, yielding a final concentration of approximately 0.054 mmol/L.

2.5 | Magnetic resonance imaging measurements

The MRI measurements were performed on three clinical 3T scanners: two 3T Prismas and one 3T Trio (Siemens Healthineers). For all Prisma measurements, the 64-channel head coil for signal reception and the body coil for transmission were used. All in vivo measurements were performed under approval by the local ethics committee. Each subject gave written, informed consent before the study. Where not specified explicitly, measurements were done on a Siemens 3T Prisma scanner at the Max Planck Institute Tuebingen). Following the *Pulseseq* SBB, a 3D gradient-echo readout,^{26,27} was used. A table with relevant imaging parameters for the readout sequence can be found in Supporting Information Table S1. In addition, all *pulseseq-files* used in this study are provided in the Supporting Information.

For the phantom experiment, a spin-lock (SL) Z-spectrum acquisition was performed using HSEXP pulses⁴³ with the pulse shape parameters recently optimized for 3 T²⁹ (see Figure 1A,B and SLEXP_3T_Phantom.seq in the Supporting Information). For each saturation, a single spin-lock pulse with a duration of 100 ms was played out after a recovery delay (T_{rec}) of 5 seconds. For this protocol, 39 evenly distributed offsets between ±6 ppm were acquired, together with an unsaturated S₀ scan. To enable realistic simulations,

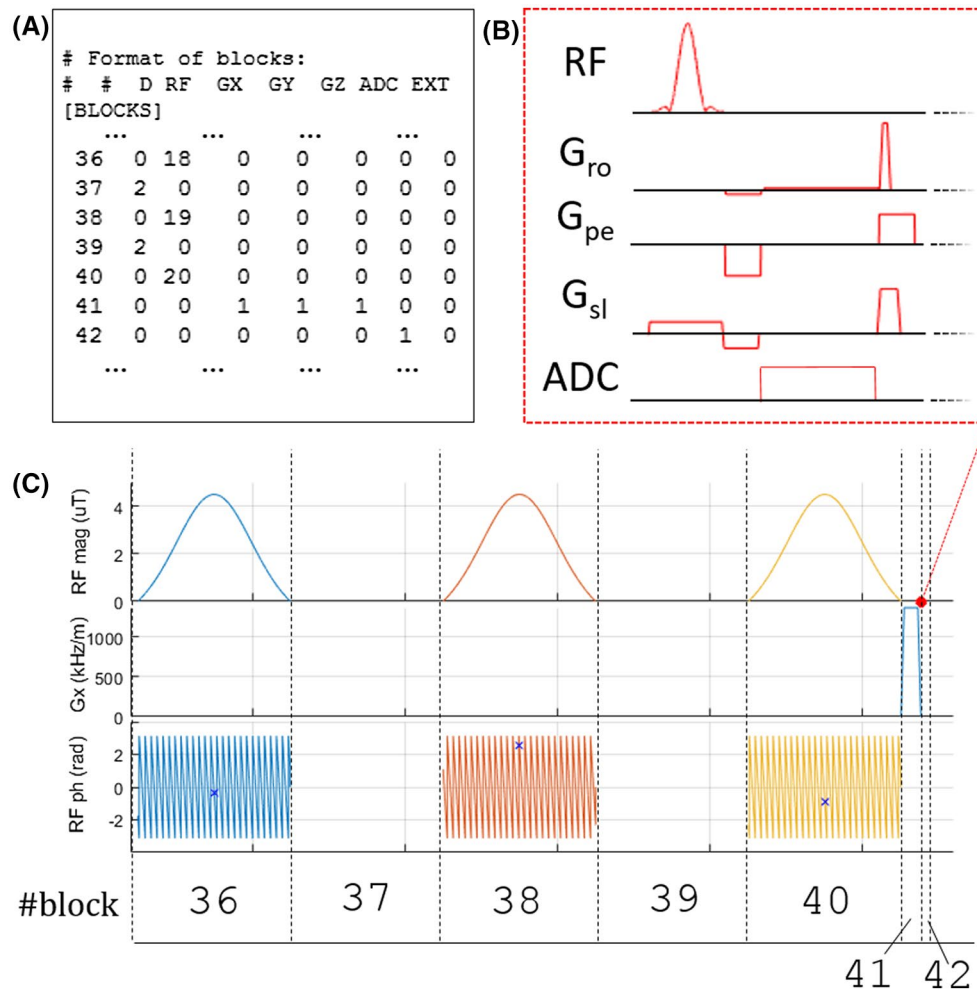


FIGURE 2 Schematic sketch of the *Pulseseq*-CEST hybrid sequence playing out the preparation period of protocol APTw_3T_002. A, Format of the *pulseseq*-file with channels for time delay, RF pulse, gradient, ADC, and trigger events. The numbers link to entries in a lookup table, where the actual event parameters are defined. Note that the RF pulses have the same amplitude and frequency offset, but a different ID due to different phase offsets. B, Example gradient-echo readout sequence. C, Proposed combination of *Pulseseq* events and the readout block using *Pulseseq* as a sequence-building block (SBB). The RF events appear at blocks number 36, 38, and 40, spoiler gradients at block number 41, and delay events at blocks number 37 and 39. All *Pulseseq*-CEST RF events are spatially nonselective. Blue crosses on the RF phase plots mark the RF phase at the moment of time when the peak RF magnitude is reached. For simplicity, GY and GZ gradient axes are not shown here. At every ADC event in the *pulseseq*-file (block 42), the gradient-echo readout is played out

a WASABI (simultaneous mapping of water shift and B_1),⁴⁴ saturation recovery, and T_2 magnetization-preparation⁴⁵ sequence were applied to determine B_0 , B_1 , T_1 , and T_2 maps. The phantom was scanned at a room temperature of about 25°C.

The three different APTw protocols shown in Figure 1C-H (APT_w_3T_001.seq, APT_w_3T_002.seq, and APT_w_3T_003.seq in the Supporting Information) were scanned for direct comparison in vivo in a healthy volunteer. In addition, a WASABI measurement was performed to correct the Z-spectra for B_0 field inhomogeneity. To generate magnetization-transfer asymmetry (MTR_{asym}) maps with higher SNR in vivo, all APTw protocols were acquired with repeated acquisitions (three repetitions) at the offsets of interest and a dummy scan at the beginning (APT_w_3T_001_AVG.

seq, APT_w_3T_002_AVG.seq, and APT_w_3T_003_AVG.seq in the Supporting Information). This averaged protocol was additionally measured in a patient with a glioblastoma (World Health Organization grade IV, IDH mutation, and methylation of MGMT (O(6)-methylguanine-DNA methyltransferase) promoter) at the University Hospital Erlangen under approval of the local ethics committee.

Finally, we performed a whole-brain, in vivo CEST-MRF³⁶ measurement at three different sites. For this purpose, the *Pulseseq* SBB was implemented into a 3D-EPI sequence.^{39,40} All relevant readout parameters can be found in Supporting Information Table S1. The protocol consisted of the following measurements: (1) an APTw MRF protocol with 31 scans, using spin-lock pulses with a B_1 varying between 0 μ T and 4 μ T at a constant offset of 3.5 ppm (Figure 1I,J); (2)

a semisolid magnetization-transfer contrast-weighted MRF protocol with 31 scans, using spin-lock pulses with B_1 varying between 0.2 μT and 4 μT at offsets varying between 6 ppm and 14 ppm; (3) a WASABI measurement for B_0 and B_1 field inhomogeneity maps; (4) a saturation-recovery measurement for T_1 maps; and (5) a T_2 preparation measurement for T_2 maps. All *pulseq-files* for this experiment can be found in the Supporting Information (MRF_Amide.seq, MRF_MT.seq, T1prep.seq, T2prep.seq, and WASABI.seq). The protocol was applied at three different sites for one healthy volunteer each (two Siemens 3T Prisma Scanners at MPI Tuebingen and Massachusetts General Hospital, and a Siemens 3T Trio scanner at University Hospital Erlangen). For the Trio system, a 32-channel coil was used for reception, whereas a 64-channel coil was used at both Prisma systems.

2.6 | Postprocessing

For the phantom experiment, MTR_{asym} maps were generated voxel-wise, with $\text{MTR}_{\text{asym}}(\Delta\omega) = (S(-\Delta\omega) - S(\Delta\omega))/S_0$, after a ΔB_0 correction using a linear interpolation between acquired offsets. Additionally, three regions of interest (ROIs) were drawn in the center slice in the tubes listed in Table 1. Z-spectra in these ROIs were simulated as described in section 2.2. All simulation parameters can be found in Supporting Information Tables S2–S4). Both simulated and measured Z-spectra were normalized by the measurement at -6 ppm.

For the in vivo APTw experiment, all measurements were motion-corrected using *elastix*.⁴⁶ The applied *elastix* parameter file can be found in the Supporting Information (Rigid_MMI.txt). The MTR_{asym} maps were generated in the same way as for the phantom but with applying an additional principal component analysis denoising approach⁴⁷ using the Malinowski criterion and a spatial 2D in-plane Gaussian filter ($\sigma = 0.6$) to smooth the images. A white-matter ROI was generated automatically by segmenting the image at -4 ppm using *SPM* (statistical parameter mapping).⁴⁸

For the CEST-MRF experiment, all images were again motion-corrected and registered to the T_1 measurement using *elastix*. The T_1 maps were generated by fitting a mono-exponential function to the data of the saturation-recovery measurement. The T_1 map was then used to generate a synthetic T_1 -weighted image, which was subsequently used to

generate gray-matter and white-matter segmentation masks using *SPM*. Dictionary generation and calculation of amide and semisolid magnetization transfer-contrast concentration and exchange-rate maps were performed according to Perlman et al.³⁸ A major advantage in using the *Pulseseq* SBB in this context is that for both measurement and dictionary generation, the same *pulseq-file* describing the saturation period definition is used. This reduces possible error sources from transferring the measurement parameters to the simulation or vice versa.

3 | RESULTS

3.1 | Simulation and phantom measurement

The first experiment demonstrates how the same *pulseq-file* can be used for simulation and phantom measurements. Figure 3 shows the MTR_{asym} map (Figure 3A) at 2 ppm, as well as the measured Z-spectra ($Z(\Delta\omega) = S(\Delta\omega)/S(-6 \text{ ppm})$) and simulated Z-spectra ($Z(\Delta\omega) = M_z(\Delta\omega)/M_z(-6 \text{ ppm})$) (Figure 3B) and the MTR_{asym} (Figure 3D) curve for the three different ROIs. The residuals between these measured and simulated normalized intensities are displayed in Figure 3C. The maximum residuals were 0.009, 0.008, and 0.004 for ROIs 1, 2, and 3, respectively. Simulation of the *pulseq-file*, using 500 samples per pulse (60 000 pulse samples per Z-spectrum) took approximately 0.19 seconds for a single amide CEST pool Z-spectrum (ROIs 1 and 2) and 0.77 seconds for four different D-glucose CEST pools (ROI 3).

3.2 | In vivo APT-weighted measurements

The second experiment demonstrates the value of well-defined *pulseq-files* for comparison of different approved APTw CEST implementations. The APTw ($\text{MTR}_{\text{asym}}(3.5 \text{ ppm})$) results for the volunteer and the tumor patient can be found in Figure 4. The MTR_{asym} maps for the three different protocols are shown in Figure 4A–C, and despite different saturation, they show similar contrast in the healthy volunteer. The contrast in healthy brain is very low, which is expected, as the APTw-imaging parameters are designed to yield almost no contrast in healthy tissue.^{4,49–51} Corresponding Z-spectra, with $Z(\Delta\omega) = S(\Delta\omega)/S(-1560 \text{ ppm})$, and MTR_{asym} spectra can be found in Figure 4H,I, respectively. Comparison with simulation can be found in Supporting Information Figure S3. While the intensity in the MTR_{asym} maps of the healthy volunteer is similar for all three protocols, a clear contrast can be seen in the tumor region (Figure 4D–F) of the glioblastoma patient. These protocols are all in use at clinical scanners, but were, to our knowledge,

TABLE 1 Region of interest and phantom information

ROI	Solute	Concentration (mmol/L)	pH	Voxels (N)
1	L-arginine	50	5.55	18
2	L-arginine	50	4.01	23
3	D-glucose	66	6.52	23

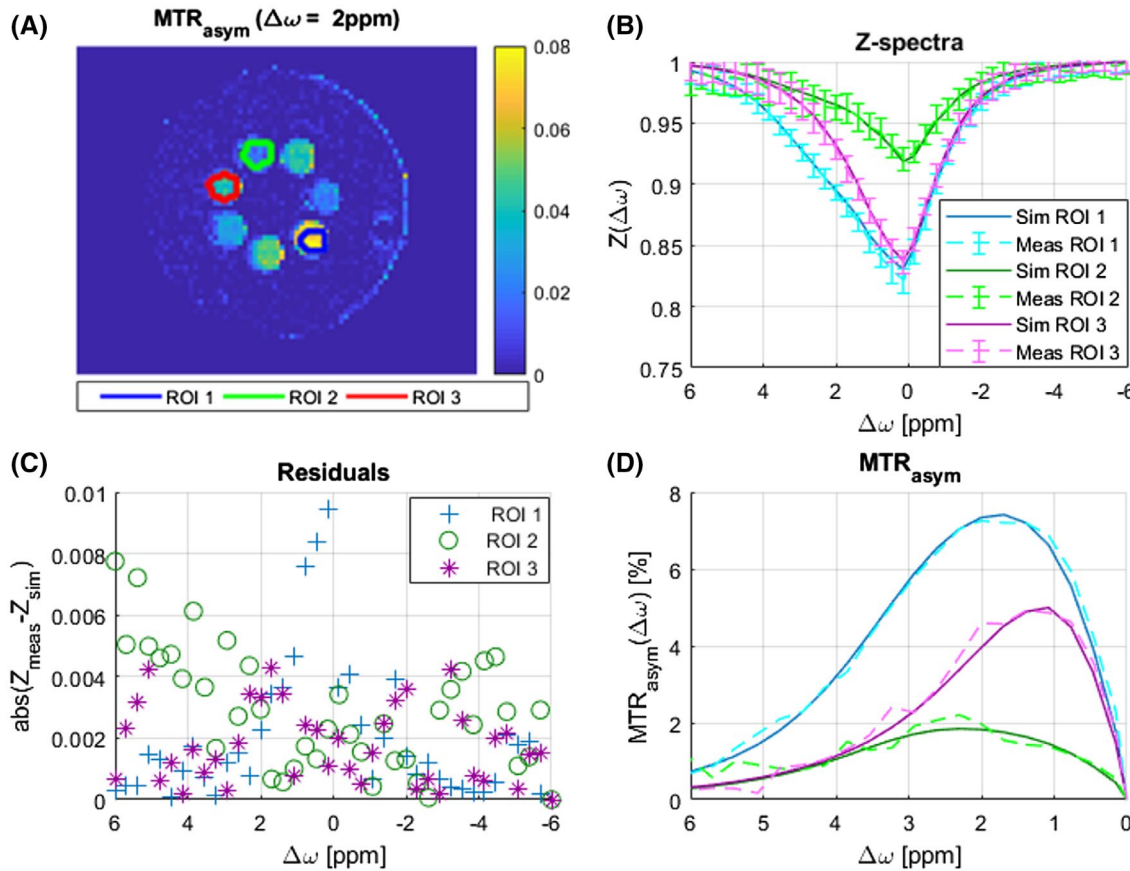


FIGURE 3 A, The $MTR_{asym}(\Delta\omega = 2\text{ppm})$ in slice 6. B, Simulated and measured Z-spectra for regions of interest (ROIs) 1, 2 and 3, respectively. Error bars in the measured Z-spectra show the SD of Z-spectra across voxels. C, Difference between the measured and simulated Z-spectra for each ROI. D, The MTR_{asym} curves of simulated and measured Z-spectra for the three ROIs

never compared side by side. Such a comparison can now be performed in different pathologies, to validate different protocols and their relation to previous work.

3.3 | In vivo CEST-MRF measurements

The final experiment demonstrates that with *Pulseseq*-CEST, sophisticated CEST sequences can be shared between sites, for both simulation (and training of reconstruction networks in this CEST-MRF example) as well as for measurement of work-in-progress developments at different systems. The spin-lock based CEST-MRF scheme with whole-brain EPI readout was applied in three healthy volunteers and at three different MR sites. The resulting maps for the semisolid proton fraction (f_{MT}), the semisolid proton to water proton exchange rate (k_{MT}), the amide proton fraction (f_{Amide}), and the amide proton to water proton exchange rate (k_{Amide}) are shown in Figure 5. The mean values for gray matter and white matter across the entire brain are shown in Table 2. A visual as well as quantitative comparison indicates reproducibility of effects across multiple sites.

4 | DISCUSSION

By adapting the source code of the *Pulseseq* interpreter sequence to the SBB concept of the Siemens IDEA framework, we were able to use it as a sequence building block in established MR sequences and subsequently run CEST experiments at different clinical 3T scanners. Hence, we combined the full flexibility of *Pulseseq* and the sophisticated readout methods from native sequences to generate an easy-to-use and flexible method for reproducible CEST measurements. For instance, it was directly possible to perform CEST-MRF experiments at three different sites on scanners with distinctly different software versions using identical CEST preparation periods defined in *Pulseseq*. Thus, a sophisticated and still work-in-progress protocol could be reliably shared between research sites using *Pulseseq*-CEST, without being limited by hard-coded user interface interactions. The quantitative results of the applied CEST-MRF method are consistent across sites, although the maps generated from the Trio system appear noisier. We attribute this primarily to the used 32-channel receive coil, leading to lower SNR compared with the 64-channel coil used on the Prisma system.

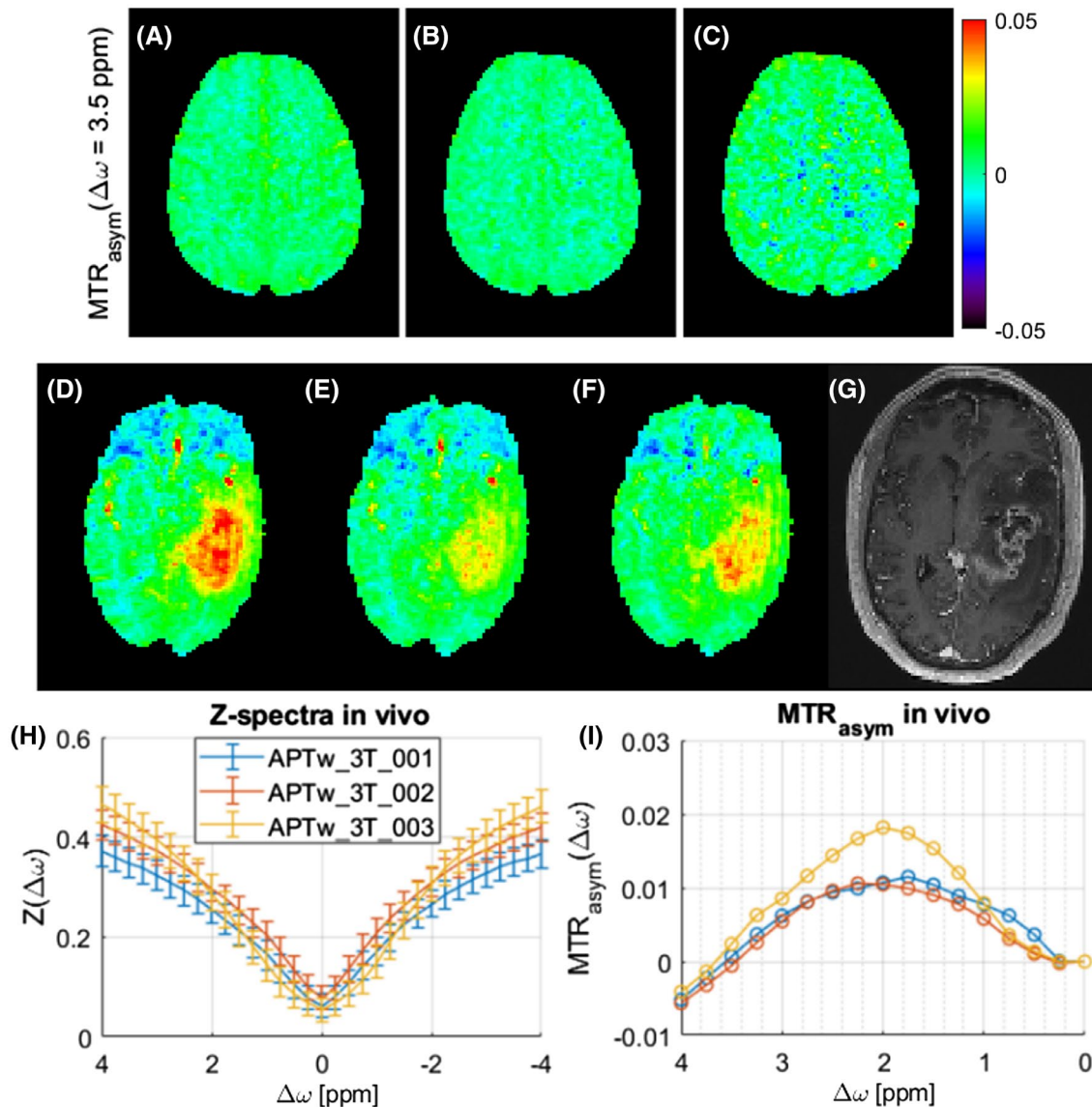


FIGURE 4 A-C, The MTR_{asym} maps at 3.5 ppm for the APTw_3T_001_AVG (A), APTw_3T_002_AVG (B), and APTw_3T_000_AVG3 (C) protocols in a healthy volunteer. D-F, The MTR_{asym} maps at 3.5 ppm for the APTw_3T_001_AVG (D), APTw_3T_002_AVG (E), and APTw_3T_003_AVG (F) protocols in a glioblastoma patient. G, Contrast-enhanced MPRAGE image for this patient. Z-spectra (H) and MTR_{asym} curve (I) for a white-matter ROI in the volunteer. Due to the rectangular saturation, residual artifacts in the ventricle regions can be observed in (C). Error bars in (H) show the SD in the white-matter ROI

In addition, the actual amplitude of the RF pulses is not only determined by the defined pulse shape in the *pulseq-file*, but also by the reference voltage of the system. It is therefore possible that different reference voltages lead to different results. Such MRF differences can now be evaluated and handled. While larger, further studies using this protocol should compare and discuss results with previous CEST-MRF studies,^{38,52} it is beyond the scope of the work presented here, as Bloch-McConnell-based quantification is extremely challenging in vivo. Results presented here represent a work-in-progress state of a sophisticated method undergoing an active development. The *Pulseseq*-CEST standard allows for efficient and active exchange between the research sites, even in such

an early stage of development, accelerating generation and refinement of simulation databases and improving model training. The *Pulseseq* SBB presented in this paper allows convenient though reliable multi-center collaboration to further investigate the method in detail in a larger cohort.

In addition, our software provides a Bloch-McConnell simulation tool for *pulseq-files* to simulate the exact same CEST preparation period that is played out by the interpreter on the scanner. The fast, native C++ implementation allows for pulsed CEST simulations, also with multiple isochromats, an application discussed in more detail in Supporting Information Figures S4 and S5. To ensure broad applicability, we provide implementations of the Bloch-McConnell simulation in both

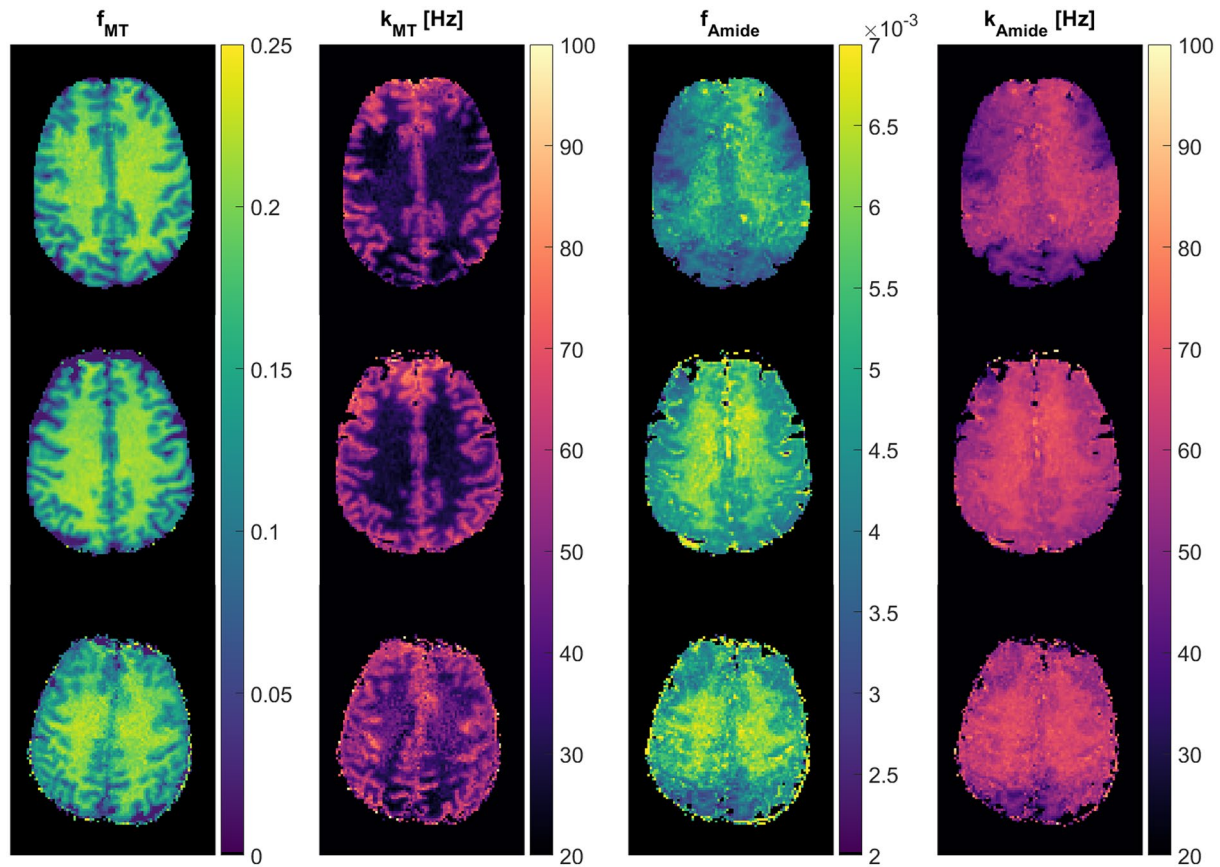


FIGURE 5 Semisolid magnetization-transfer (MT) contrast water fraction (first column) and exchange rate (second column), amide water fraction (third column), and exchange rate (fourth column) for measurements at a 3T Prisma in Tuebingen (first row), 3T Prisma in Boston (second row), and 3T Trio in Erlangen (third row)

MATLAB and *Python*, which are the most frequently used programming languages in research. Confirmation that both implementations yield identical results is provided in Supporting Information Table S5. We hope that with this work, we provide the first version of a valuable and needed tool for the CEST community, to exchange and test CEST preparation periods for the many different types of different CEST experiments. For instance, a researcher publishing data could share a *pulseseq-file* containing all RF pulse, gradient, and delay parameters through the Supporting Information, and it could subsequently be used by other researchers. Additionally, all *pulseseq-files* can be made available on the project's website (<https://pulseseq-cest.github.io/>).

Moreover, the *Pulseseq* sequence building block can be used to test all of these CEST-preparation blocks with a minimum workload, even with different readout sequences, such as gradient echo, EPI or RARE, which have been optimized with regard to their imaging performance beforehand. Although the SBB is so far only available for Siemens scanners, it already demonstrated its multiplatform capabilities by executing the same CEST preparation periods on scanners built on different hardware components and running different software versions. Furthermore, it is generally possible

to transfer the approach to GE and Bruker (Bruker Biospin, Ettlingen, Germany) systems, where *Pulseseq* implementations have been demonstrated.²³ For Bruker systems, we recently proposed an initial approach to combine CEST preparation periods from *pulseseq-files* with native Bruker readouts automatically with *MATLAB*.⁵³ By doing so, we were able to measure the above-mentioned APTw protocols on a Bruker 14.1T scanner with *ParaVision* 6. This will be useful for comparison between preclinical and clinical trials, especially for pulsed CEST approaches.

A design of the interpreter software for other manufacturers (eg, Philips, United Imaging, or Canon) is needed for a universal application. However, even without actually applying *pulseseq-files* on the scanner, the possibility to share, edit, display, and simulate saturation periods can be very insightful and beneficial for the design of CEST experiments. Because the *pulseseq-file* guarantees a completely defined CEST preparation period inherently, which we believe is needed to improve the reproducibility of data in the CEST community, this will be a valuable tool especially when designing multisite clinical trials of technology, such as APTw MRI of brain tumors, which is being performed currently at many sites but often with different protocols.

TABLE 2 Mean values for quantitative CEST-MRF parameters and relative B_1 values of gray matter and white matter at the three different sites (Tuebingen, Boston, and Erlangen)

	f_{MT} GM	k_{MT} GM (Hz)	f_{MT} WM	k_{MT} WM (Hz)	f_{Amide} GM	k_{Amide} GM (Hz)	f_{Amide} WM	k_{Amide} WM (Hz)	Rel. B_1 (GM)	Rel. B_1 (WM)
TUE Prisma	0.140 ± 0.033	47.3 ± 10.8	0.201 ± 0.017	31.7 ± 6.7	0.0046 ± 0.0010	54.6 ± 10.0	0.0052 ± 0.0009	60.7 ± 7.5	1.03 ± 0.10	1.05 ± 0.09
BOS Prisma	0.142 ± 0.038	47.7 ± 11.3	0.203 ± 0.019	32.0 ± 6.6	0.0049 ± 0.0010	58.3 ± 8.8	0.0055 ± 0.0008	63.2 ± 6.0	0.99 ± 0.09	1.02 ± 0.08
ERL Trio	0.142 ± 0.034	48.6 ± 11.2	0.197 ± 0.022	35.2 ± 8.6	0.0047 ± 0.0011	55.8 ± 10.3	0.0054 ± 0.0009	61.9 ± 7.3	1.01 ± 0.10	1.03 ± 0.09

Abbreviations: BOS, Boston; ERL, Erlangen; GM, gray matter; TUE, Tuebingen; WM, white matter.

The *Pulseseq* interpreter sequence has been adapted and used as a sequence building block in native readout sequences herein, but it is also possible to use readouts implemented directly in *Pulseseq*. The current version of the *Pulseseq* interpreter allows for the images to be reconstructed directly on the scanner, which makes the development of novel imaging sequences even more convenient. As the presented approach is compatible with the “parent” *Pulseseq* project, the previously published *Pulseseq*-CEST preparation periods could be trivially integrated with these novel readout modules, implemented directly in *Pulseseq*. This will allow full reproducibility, as the complete sequence (including preparation and readout) can be published in the *pulseseq-file* format in the same database.

While native *Pulseseq* readout modules can be considered for measurements in the future, it is already possible to combine them with CEST preparation modules for more realistic simulations. In that case, the pseudo ADC event needs to be replaced by a full readout sequence. Various examples of possible readout sequences implemented with *Pulseseq* can be found on the project’s website (<https://pulseseq.github.io/>). By explicitly including the readout, it becomes possible to simulate the influence of the RF pulses during the readout with the provided simulation (Supporting Information Figure S6). It is also possible to use a different *Pulseseq*-compatible simulation software, such as *JEMRIS*,⁵⁴ which supports Bloch-McConnell simulations, and provides a full MRI simulation framework. In addition, if the readout module is included in the *pulseseq-file*, it is possible to estimate the expected specific absorption rate value using *sar4seq*⁵⁵ before the scanning session. In general, the open format of *Pulseseq* allows for a broad development of useful applications.

With regard to the growing number of applications of neural networks to CEST data reconstruction,^{22,38,56-58} often trained on simulated data, a match of the sequences used in silico and in vivo becomes crucial. In fact, the CEST-MRF reconstruction network used to infer the quantitative maps of Figure 5 was trained using simulated data with exactly the same *pulseseq-file* as used at the MR scanner. This is just one example showing that *Pulseseq*-CEST could be valuable for many emerging machine and deep learning-based approaches.

Finally, the method is obviously not limited to CEST applications. In principle, any magnetization-preparation sequence can be realized, as exemplified in Supporting Information Figure S7, for the WASABI field-mapping approach and a T_1 saturation-recovery sequence.

5 | CONCLUSIONS

We present a *Pulseseq*-based sequence framework for CEST preparation pulse sequences and an accompanying simulation tool, the use of which in combination with available MRI sequences allows for straightforward sharing, implementing,

testing, optimizing, and running of CEST MRI studies. Because the *pulseseq-files* inherently include a complete CEST parameter definition, this fosters faster comparison and facilitates reproducibility—not only between different MR sites, but also between real and simulated environments.

Source code for the Siemens IDEA interpreter sequence is available on request. All code for creating and simulating *pulseseq-files* is open source and can be obtained on the project's website (<https://pulseseq-cest.github.io/>).

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DATA AVAILABILITY STATEMENT

All source code used for *pulseseq-file* generation and simulation in this study is openly available at <https://pulseseq-cest.github.io/>, where links to the corresponding GitHub repositories can be found.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the Supporting Information section.

FIGURE S1 Single magnetization vector trajectory during the CEST preparation period at 3.5 ppm with phase accumulation transfer during the saturation pulses (A,B) and without (C,D). The eight different colors indicate the trajectory during the eight different pulses. Due to the phase difference

between magnetization vector and RF pulse in (C) and (D), large oscillations can occur

FIGURE S2 Z-spectra of a single magnetization vector for the sequences from Supporting Figure 1A,B (blue) and C,D (red). Depending on the frequency offset $\Delta\omega$, the artifacts are more or less severe, as the accumulated phase and therefore the phase difference between magnetization vector and RF pulse is dependent on the frequency offset and the duration of the pulse

FIGURE S3 Measured (A,B) and simulated (C,D) Z-spectra (A,C) and magnetization-transfer asymmetry (MTR_{asym}) curves (B,D) for the three different amide proton transfer-weighted (APT_w) protocols from the main paper

FIGURE S4 Comparison of simulation results between a single (blue) and multiple isochromats (red) in different tissue environments. The red curve shows the mean Z-spectrum of all 200 isochromats, and the error bars indicate the SD between isochromats

FIGURE S5 Trajectory during a rectangular saturation pulse of a single isochromat with short (left) and long T_2 (right)

FIGURE S6 Evolution of the z-magnetization after different CEST-preparation periods at 3.5 ppm during a 2D gradient-echo readout

FIGURE S7 Comparison of simulation results from the *MATLAB* implementation (solid blue) and the *Python* implementation (dashed orange) for a WASABI (A) and a T_1 saturation-recovery (B) example

TABLE S1 Basic imaging parameters of readouts used for the presented Pulseseq-hybrid sequence

TABLE S2 Simulation parameters for first region of interest (ROI)

TABLE S3 Simulation parameters for second ROI

TABLE S4 Simulation parameters for third ROI

TABLE S5 Currently available pulseseq-files in the Pulseseq-CEST open database with the RMS error of the spectra simulated in *MATLAB* and *Python*

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